

Einstein's Mass-Energy Relation



MD. GULAM MUSTAFA

Lecturer (Physics)

Department of Science & Humanities

Bangladesh Army International University of Science and Technology (BAIUST)

Syedpur, Cumilla-3501

Whatever we will discuss

- Brief discussion on Mass Defect and Binding Energy
- Deduction of Einstein's Mass-Energy Relation
- Mathematical problems related to Mass-Energy Relation

Related Terms

Mass Defect

Binding Energy

Mass defect

- The mass of the permanent nucleus is found to be less than the combined mass of the constituent elements.

Binding Energy

- The amount of energy in order to form a permanent nucleus by combining the necessary number of nucleons.

Mass-Energy Relation

Deduction of Einstein's Mass-Energy relation,

$$E = mc^2$$

It's the world's most famous equation, but what does it really mean? "Energy equals mass times the speed of light squared." On the most basic level, the equation says that energy and mass (matter) are interchangeable; they are different forms of the same thing.

Let us suppose that force F is applied to an object to bring it to its state of motion and the object covers a distance ds . The kinetic energy is Fds . If the object covers the total distance s then the total kinetic energy

$$T = \int_0^s F ds$$

We know that

$$F = \frac{d}{dt}(mv)$$

$$\text{And } \frac{ds}{dt} = v \therefore ds = v dt$$

Now if we put the value of F and ds we get,

$$T = \int_0^{mv} \frac{d}{dt}(mv) v dt$$

$$= \int_0^{mv} v d(mv)$$

$$= \int_0^{mv} v(v dm + m dv) \quad \left[\because \text{When } s = 0 \text{ then } mv = 0 \text{ and } \right. \\ \left. \text{when } s = s, \text{ then } mv = mv \right]$$

$$= \int_0^{mv} (v^2 dm + m v dv) \quad \dots\dots\dots (1)$$

From the relativity of mass we know that

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}} \text{ . By squaring this equation we get,}$$

$$m^2 = \frac{m_0^2}{1 - v^2/c^2}$$

$$\text{or, } \left(1 - \frac{v^2}{c^2}\right) m^2 = m_0^2$$

$$\text{or, } \left(\frac{c^2 - v^2}{c^2}\right) m^2 = m_0^2$$

$$\text{or, } m^2 c^2 - m^2 v^2 = m_0^2 c^2$$

By differentiating this equation we get

$$2mc^2 dm - (2mv^2 dm + 2vm^2 dv) = 0$$

Dividing the above equation by $2m$ we get,

$$c^2 dm - v^2 dm = mv dv$$

Putting the value of $mv dv$ in equation (1) we get

$$T = \int_{m_0}^m c^2 dm \left[\begin{array}{l} \because \text{When } mv = 0 \text{ then } m = m_0 \text{ and} \\ \text{when } mv = mv \text{ then } m = m \end{array} \right]$$

Here m_0 is the rest mass,

$$\therefore T = c^2 \int_{m_0}^m dm$$

$$= c^2 [m]_{m_0}^m$$

$$= c^2 (m - m_0)$$

$$= mc^2 - m_0 c^2$$

$$\text{or, } T + m_0 c^2 = mc^2$$

But total energy $E = T + m_0 c^2 = \text{kinetic energy} + \text{potential energy}$

$$\therefore E = mc^2$$

Related Mathematical Problems

❖ A particle with mass 9.1×10^{-28} kg is totally converted to energy. What will be the amount of energy?

[Ans. 8.19×10^{-11} J]

❖ Determine the equivalent energy of 1 gm mass in Joule and in MeV.

[Ans. 9×10^{13} J and 5.625×10^{26} MeV]

❖ Determine the equivalent energy of an electron mass in MeV.

[Ans. 0.51 MeV]

❖ Express the equivalent energy of 12 a.m.u. in eV.

[Ans. 1.12×10^{10} eV]



Thank You
Thanks for your attention